

SYNOPSIS

On 11 February 2015, the United Kingdom registered container ship *Ever Smart* collided with the Marshall Islands registered oil tanker *Alexandra 1* near the entrance to the buoyed approach channel in Jebel Ali, United Arab Emirates. The container ship was outbound at a speed of 12 knots and had disembarked its pilot. The tanker was inbound and was moving very slowly ahead while waiting for the pilot from the container ship to board. Both vessels suffered major structural damage to their bows but there were no injuries or pollution.

The collision resulted from several factors. In particular, a passing arrangement was not agreed or promulgated and the actions of both masters were based on assumptions. *Alexandra 1* was unnecessarily close to the channel entrance and the tanker's master acted on scanty VHF radio information. In addition, *Ever Smart*'s bridge team did not keep a proper lookout or monitor the tanker's movement. They only realised that *Alexandra 1* was close ahead seconds before the collision when alerted by the port control.

The accident occurred within Jebel Ali's port limits. The precautions of pilotage and the port's vessel traffic service, which would normally co-ordinate and de-conflict the movements of vessels in the port area, were ineffective on this occasion.

Evergreen Marine (UK) Limited, the managers of *Ever Smart* and Iships Management Private Limited, the managers of *Alexandra 1* have taken action to improve the standard of bridge watchkeeping on board their vessels. A recommendation to DP World UAE Region, the operators of Jebel Ali port, is intended to improve the effectiveness of the vessel traffic and pilotage services it provides.

SECTION 1 – FACTUAL INFORMATION

1.1 PARTICULARS OF *EVER SMART*, *ALEXANDRA 1* AND ACCIDENT

SHIP PARTICULARS		
Vessel's name	<i>Ever Smart</i>	<i>Alexandra 1</i>
Flag	United Kingdom	Marshall Islands
Classification society	Lloyd's Register	Det Norske Veritas
IMO number	9300403	9127148
Type	Container ship	Crude oil tanker
Registered owner	Evergreen Marine (UK) Limited	Nautical Challenge Limited
Manager	Evergreen Marine (UK) Limited	Iships Management Private Limited
Construction	Steel	Steel
Year of build	2005	1997
Length overall	299.99m	269.19m
Gross tonnage	75246	79779
Minimum safe manning	16	20
Authorised cargo	Containers	Oil in bulk
VOYAGE PARTICULARS		
Port of departure	Jebel Ali, United Arab Emirates	Umm Al Qasr, Iraq
Port of arrival	Dammam, Saudi Arabia	Jebel Ali, United Arab Emirates
Type of voyage	International	International
Cargo information	48564t in containers	113973.5t condensate
Draft	12.7m	14.0m
Manning	21	28

MARINE CASUALTY INFORMATION		
Date and time	11 February 2015 1942 UTC	
Type of marine casualty or incident	Serious Marine Casualty	
Location of incident	Jebel Ali, United Arab Emirates	
Vessel's name	<i>Ever Smart</i>	<i>Alexandra 1</i>
Place on board	Bow/forecastle deck	Bow/forecastle deck
Injuries/fatalities	None	None
Damage/environmental impact	Bow severely distorted and penetrated.	Bow and forecastle deck severely damaged.
Ship operation	On passage	On passage
Voyage segment	Transit	Transit
External & internal environment	Wind: 068° at 6 knots. It was dark with clear skies and good visibility. The sea was calm with a low swell. The tidal stream was south-easterly at less than 1 knot.	
Persons on board	21	28

1.2 NARRATIVE

1.2.1 Events leading up to the collision

During the evening of 11 February 2015, the United Kingdom (UK) registered container ship *Ever Smart* was alongside in Jebel Ali, United Arab Emirates (UAE). At 2128, the vessel's third officer contacted Jebel Ali port control by very high frequency (VHF) radio channel 69¹ and advised that cargo operations would be completed in 20 minutes. The duty vessel traffic service officer (VTSO) acknowledged the call. Details of the vessel's next port of call, pilot boarding arrangements and the number of tugs to be used for departure were exchanged.

At 2200, the inbound Marshall Islands registered oil tanker *Alexandra 1*, loaded with 113973.5t of condensate², anchored in the short-term anchorage off Jebel Ali (**Figure 1**) to wait for a pilot. At the same time, the VTSO called the tanker and passed instructions for the pilot's embarkation (**Table 1**).

VTSO	<i>Alexandra 1 the pilot will board at about 2315 by buoy No1</i>
<i>Alexandra 1 (master)</i>	<i>Jebel Ali Port control this is Alexandra 1. We are now anchored, 2200.</i>
VTSO	<i>Alexandra 1 this is Jebel Ali Port control. 2315 at buoy No1</i>
<i>Alexandra 1 (master)</i>	<i>Roger. 2315 buoy No1. Shokran</i>

Table 1 – VHF radio exchange between Jebel Ali port control and *Alexandra 1* at 2200

At 2206, *Ever Smart* finished cargo operations. A pilot embarked and went to the bridge where the third officer was testing the vessel's engine in readiness for departure. The third officer and the pilot were soon joined on the bridge by the master.

The pilot was given the vessel's pilot card, which detailed its manoeuvring characteristics. The master and pilot then discussed the departure plan. The discussion included the arrangements for the pilot's disembarkation in the designated pilot boarding area (**Figure 1**). On completion, the pilot called Jebel Ali port control and was given permission for *Ever Smart* to sail.

At 2230, *Ever Smart* was pulled off its berth by three harbour tugs. At 2242, the tugs were released and the vessel was manoeuvred towards the main channel. Hand steering was selected; an able seaman (AB) was at the helm.

At 2248 *Alexandra 1*'s master reported to Jebel Ali port control that the tanker had weighed anchor and was underway. The VTSO replied:

"Proceed to buoy No1. Keep watch all of the time".

Three minutes later, *Alexandra 1*'s engine telegraph was set to 'half ahead' and the tanker moved towards the entrance to the buoyed channel (**Figure 1**). The tanker was displaying navigation lights appropriate to its length and a local signal,

¹ Unless otherwise stated, all VHF communications referred to in this report were on channel 69.

² Condensate – a light oil that is a low-density mixture of hydrocarbon liquids having gaseous components which remain as liquid under normal temperatures and pressures.

a flashing red light at the masthead, to indicate it was carrying a dangerous cargo. *Alexandra 1*'s master was accompanied on the bridge by the third officer and an AB. The vessel was in 'hand steering'; the AB was at the helm.

At 2253, the VTSO informed *Ever Smart*'s pilot via VHF radio of two inbound vessels. He also asked him to board *Alexandra 1* and pilot the tanker to its allocated berth. The pilot acknowledged the VTSO's request and advised that *Ever Smart* would remain in the channel until clear of the No1 buoys (**Table 2**).

VTSO	<i>[pilot's name] this is Jebel Ali port control. One vessel approaching C buoy. Second vessel by buoy 6 – Viking Emerald.</i>
Pilot	<i>Jebel Ali port control this is [pilot's name]. One vessel inbound approaching C buoy. Port to port. Second vessel by buoy 6 – Viking Emerald.</i>
VTSO	<i>Jebel Ali port control [pilot's name] Please board Alexandra 1. Draught 14m. Board to 9A.</i>
Pilot	<i>9A okay. My ship Ever Smart will continue to buoy No1.</i>

Table 2 – VHF radio exchange between Jebel Ali port control and the pilot at 2253

The VTSO then immediately called *Alexandra 1* and he advised that the tanker's pilot was on board *Ever Smart*. He also authorised the tanker to enter the channel as soon as the container ship was clear of the No1 buoys (**Table 3**).

VTSO	<i>Alexandra 1 Jebel Ali port. Captain, your pilot is on the outbound Ever Smart, passing buoy 12 and the Ever Smart will continue to buoy No1. They will be doing a good speed. Once Ever Smart is clear then you can enter the channel.</i>
<i>Alexandra 1 (Master)</i>	<i>Roger, sir, copy that. Coming close 1nm to buoy No1. Will wait until other vessel leave the channel thank you.</i>

Table 3 – VHF radio exchange between Jebel Ali port control and *Alexandra 1* at 2256

At 2302 *Ever Smart* passed the inbound cargo vessel *Trinityborg*. Seven minutes later, the container ship passed the inbound car carrier *Viking Emerald* (**Figure 2**). *Ever Smart* was making good a speed over the ground (SOG)³ of 12 knots (kts).

At 2312, *Alexandra 1*'s master saw by radar that *Ever Smart* was passing No8 buoys. He selected the container ship's radar target using the automatic radar plotting aid (ARPA), which enabled him to monitor its progress. The master realised that *Ever Smart* would not be clear of the channel for some time and at 2314 he set the engine telegraph from 'dead slow ahead' to 'stop'. The tanker was drifting on an easterly heading 1.3nm from No1 buoys and the master was frustrated at being off the channel entrance earlier than was necessary.

At 2319, *Alexandra 1*'s master called Jebel Ali port control and confirmed the requirements for the pilot ladder. By now, the tanker was 1.058nm from No1 buoys, making good a course over the ground of 126° at 2.2kts.

At 2328, *Alexandra 1*'s engine telegraph was set to 'dead slow ahead'. One minute later, the tug *Zakheer Bravo* called Jebel Ali port control by VHF radio and requested permission to cross the pilot embarkation area. The tug and its tow were 1.3nm to

³ All speeds are SOG unless otherwise stated.

Image courtesy of Jebel Ali Port



Figure 2: VTS radar showing *Ever Smart*'s position at 2309

the west of No1 buoys (**Figure 3**) and were on passage to Jumeirah to the east of Jebel Ali. The VTSO asked the tug's skipper "*can you see the big tanker waiting?*" The tug's skipper advised that he could and the VTSO instructed him to "*cross 1nm astern of the tanker*".

Alexandra 1's master heard part of this radio exchange and assumed that Jebel Ali port control was talking to *Ever Smart*. The master assessed that in order to pass astern of his vessel, *Ever Smart* would alter course to port on clearing the channel.

At 2331, as *Ever Smart* was approaching No3 buoys, the pilot and the master discussed the pilot's disembarkation (**Table 4**). At 2332, *Alexandra 1*'s engine telegraph was set to 'stop'. The tanker was 7.7 cables from the channel entrance at a speed of 1.8kts and maintaining a heading of 100°.

Pilot	<i>So captain, the time has come for me to go. Just follow the channel</i>
Master	<i>Do you think I can go myself?</i>
Pilot	<i>Yes, yes. There is this coming now. There is just the one ship. Only this tanker</i>
Master	<i>Yes, yes</i>
Pilot	<i>It's coming. It will wait.... Anyway I go there beforeokay captain</i>
Master	<i>Yes, yes</i>

Table 4 – Conversation between the pilot and *Ever Smart*'s master at 2331

Image courtesy of Jebel Ali Port

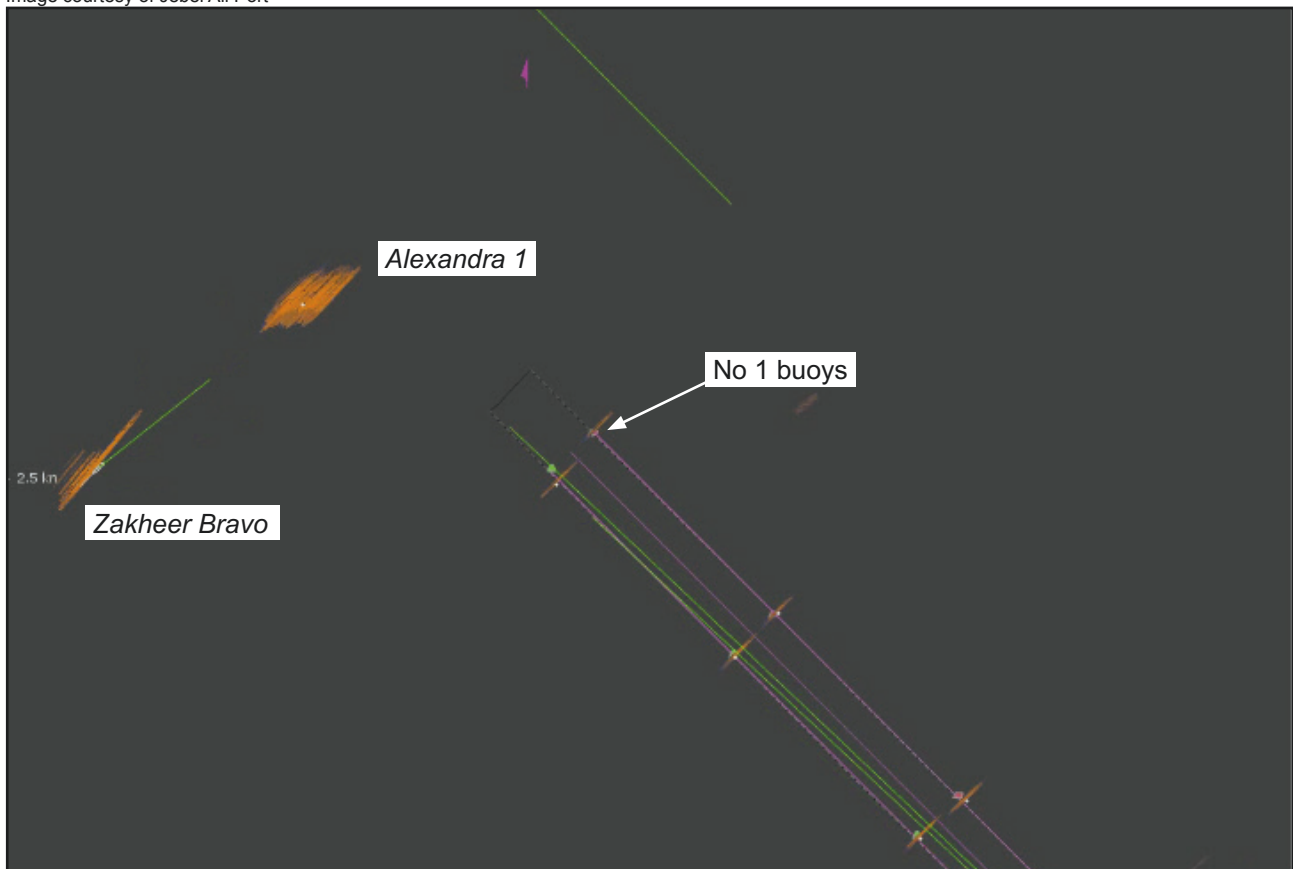


Figure 3: VTS radar showing *Zakheer Bravo* and *Alexandra 1* at 2329

Alexandra 1 was visible from *Ever Smart*'s bridge. The tanker was also on the radar displays but it was not acquired as an ARPA target (**Figure 4**).

At 2334 *Alexandra 1*'s engine telegraph was again set to 'dead slow ahead'. At about the same time, the pilot advised *Ever Smart*'s master to reduce speed to 10kts and to maintain a course over the ground of 314°. He also reminded the master of the tanker waiting to the west of the entrance to the buoyed channel; *Alexandra 1* was 0.7nm from the No1 buoys. The pilot then left the bridge, accompanied by the third officer. The master ordered the helmsman to steer 319° and adjusted the port radar display (**Figure 5**) to 'north-up' (**Figure 6**). By eye, he estimated that the tanker would pass down the container ship's port side at a distance of 1.5 cables.

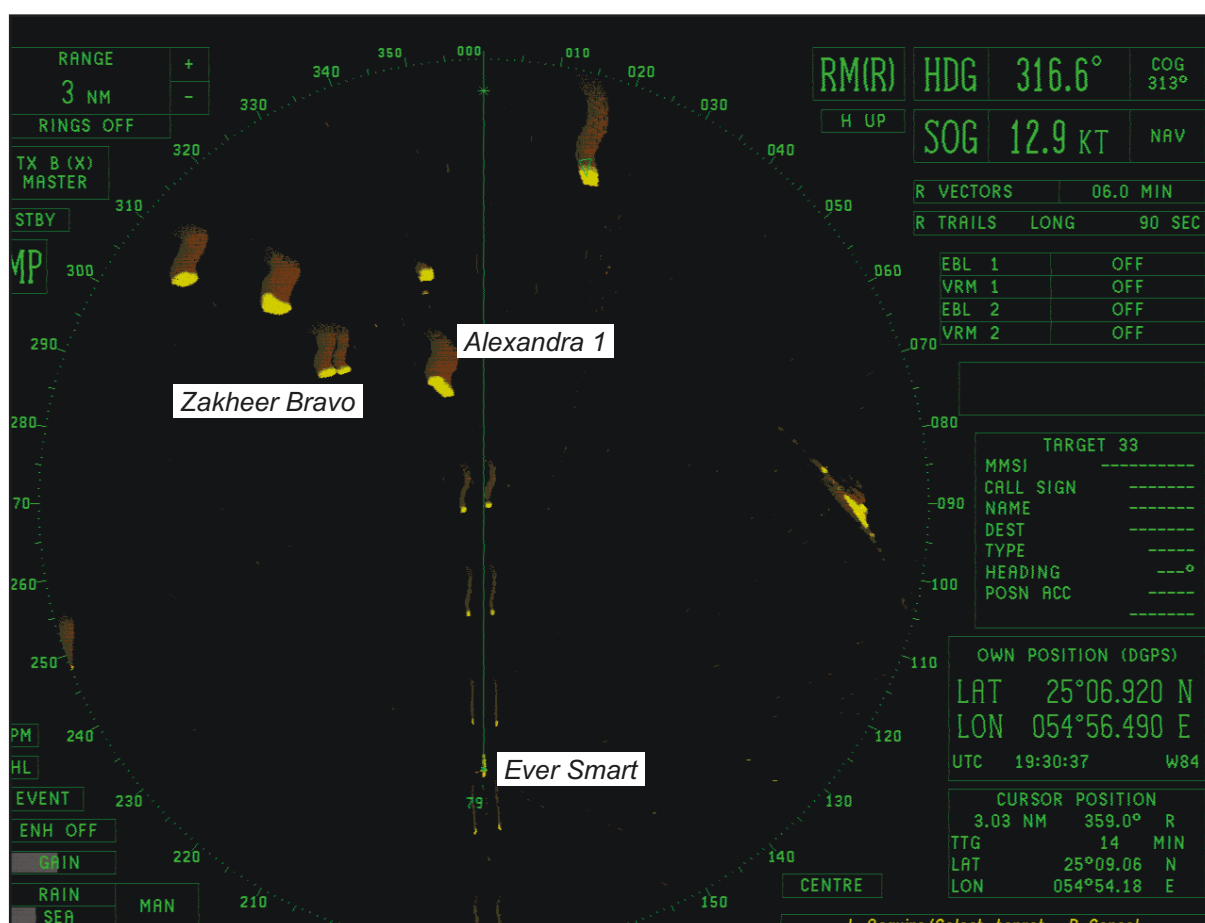


Figure 4: *Ever Smart* – port radar display at 2331

At 2337, as *Ever Smart* passed between the No2 buoys (**Figure 7**), the pilot launch with the pilot on board cleared the container vessel's port side and headed towards *Alexandra 1*. *Ever Smart*'s master immediately increased the vessel's engine speed. At the same time, *Alexandra 1*'s master confirmed to the pilot launch via VHF radio that the pilot ladder was rigged on the tanker's starboard side.

1.2.2 The collision

At 2340, *Alexandra 1*'s master moved the engine telegraph from 'dead slow ahead' to 'slow ahead'; the tanker's speed was about 2kts and it remained on an easterly heading. Approximately 30 seconds later, *Ever Smart* passed between the No1 buoys at a speed of 11kts (**Figure 8**). The container ship's third officer returned to the bridge and, on instruction from the master, he rang 'full away' with an engine setting of 80rpm. He then started to collect information required to fill in the deck log. The helmsman continued to steer 319°.

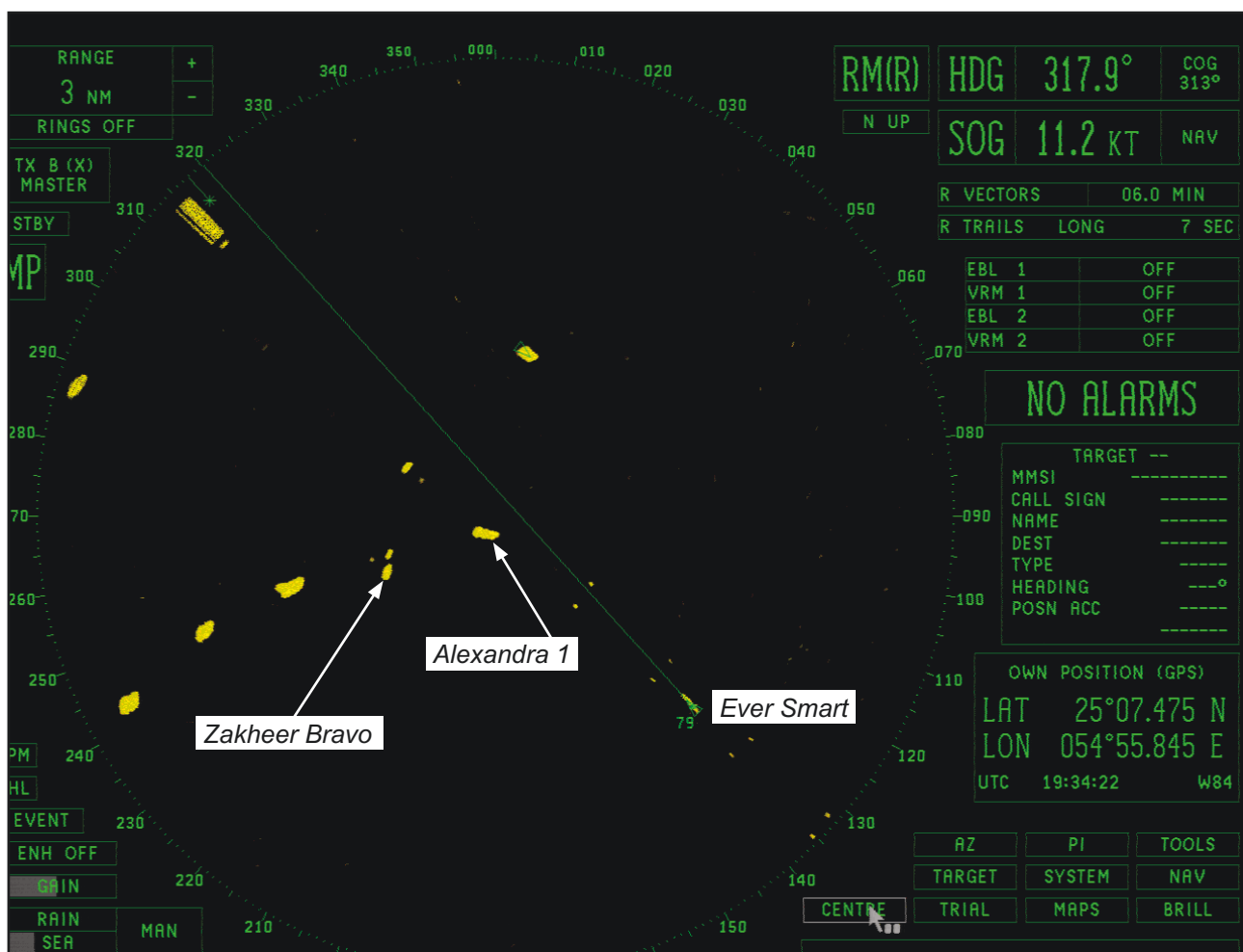


Figure 6: *Ever Smart* – port radar display at 2334

Alexandra 1's master saw *Ever Smart* pass between the No1 buoys and became concerned that the container ship had not altered course to port as he had expected. At 2341:28, *Alexandra 1*'s master called Jebel Ali port control on the VHF radio (**Table 5**).

2341:28	<i>Alexandra 1</i> (Master)	<i>Jebel Ali port control this is Alexandra 1 come in. Container not changing course. This is collision</i>
	VTSO	<i>I told him. Are they clear of buoy No1 then you will be entering the channel I said</i>
	<i>Alexandra 1</i> (Master)	<i>He's going to collision to me now!</i>

Table 5 – VHF radio exchange between *Alexandra 1* and Jebel Ali port control at 2341

The VTSO immediately called *Ever Smart*. The pilot, who was still on board the pilot launch and *Alexandra 1*'s master also intervened (**Table 6**).

2341:48	VTSO	<i>Ever Smart this is Jebel Ali port</i>
2341:52	<i>Ever Smart</i> (third officer)	<i>Jebel Ali port this is Ever Smart. Good morning ...</i>
2341:55	VTSO	<i>Are you clearing to starboard please? We have the tanker there coming to enter the channel... [overspoken]</i>
2341:55	Pilot	<i>Ever Smart, Hard to starboard! Hard to starboard! Hard to starboard!</i>
2342	<i>Alexandra 1</i> (master)	<i>Hard to ***** starboard Hard to starboard. Ever Smart hard to starboard.</i>

Table 6 – VHF radio exchange between Jebel Ali port control, *Ever Smart*, the pilot and *Alexandra 1* at 2341 (continued)

During these VHF transmissions, *Alexandra 1*'s engine telegraph was set to 'full astern'; the tanker's deck lights and external accommodation lights were also switched on. At 2342:12, *Ever Smart*'s master ordered 'OK hard to starboard'. He then exclaimed "what's that?" Three seconds later, at 2342:19, *Ever Smart* and *Alexandra 1* collided bow to bow (**Figures 9, 10 and 11**). The vessels were 4 cables from the No1 buoys.



Figure 9: *Ever Smart* – port radar display at 2342:07

1.6.7 Previous accident

On 10 February 2009, the inbound Malta registered tanker *Kashmir* and the outbound Singapore registered container vessel *Sima Saman* collided in the vicinity of the No6 buoys in the Jebel Ali approach channel. It was daylight but the visibility was occasionally reduced to about 200m in fog. Both vessels had pilots on board.

A port to port passing had been agreed but *Kashmir* was turned across the channel in order to avoid the green (western) No6 buoy. Once the vessel was clear of the buoy, starboard helm was used to manoeuvre the vessel back towards the western side of the channel. However, collision with *Sima Saman*, which was in the centre of the channel, could not be avoided. *Sima Saman*'s bow penetrated *Kashmir*'s hull in the vicinity of the tanker's port manifold. The subsequent fire and explosion (**Figure 14**) caused severe damage to both vessels. Two of the tanker's crew were injured. There was no pollution.

Following the accident, among other things, DP World UAE introduced measures to restrict the passing of certain types and sizes of vessel in the channel (see paragraph 1.6.5).



Figure 14: *Kashmir* on fire in the Jebel Ali approach channel

1.8 AUTOMATIC IDENTIFICATION SYSTEM

1.8.1 Carriage

The International Convention for the Safety of Life at Sea 1974, as amended (SOLAS), requires all cargo ships of 300gt and over that are engaged on international voyages to be fitted with AIS. Ships fitted with AIS are required to maintain AIS in operation at all times except where international agreements, rules or standards provide for the protection of navigational information.

1.8.2 Guidance

The IMO Resolution A.917 (22), which provides guidance on the use of AIS, includes:

INHERENT LIMITATIONS OF AIS

31. The officer of the watch should always be aware that other ships, in particular leisure craft, fishing boats and warships, and some coastal shore stations including Vessel Traffic Service centres, might not be fitted with AIS.

32. The OOW should always be aware that other ships fitted with AIS as a mandatory carriage requirement might switch off AIS under certain circumstances by professional judgement of the master.

USE OF AIS IN COLLISION AVOIDANCE SITUATIONS

39. The potential of AIS as an anti collision device is recognised and AIS may be recommended as such a device in due time.

40. Nevertheless, AIS information may be used to assist collision avoidance decision making. When using the AIS in the ship to ship mode for anti collision purposes, the following precautionary points should be borne in mind: a. AIS is an additional source of navigational information. It does not replace, but supports, navigational systems such as radar target tracking and VTS; and b. The use of AIS does not negate the responsibility of the OOW to comply at all times with the Collision Regulations

41. The user should not rely on AIS as the sole information system, but should make use of all safety relevant information available

43. Once a ship has been detected, AIS can assist tracking it as a target. By monitoring the information broadcast by that target, its actions can also be monitored. Changes in heading and course are, for example, immediately apparent, and many of the problems common to tracking targets by radar, namely clutter, target swap as ships pass close by and target loss following a fast manoeuvre, do not affect AIS. AIS can also assist in the identification of targets, by name or call sign and by ship type and navigational status. [sic]

The IMO guidelines for the use of AIS have been included in Marine Guidance Note 324 (M+F) – Radio: Operational Guidance on the Use of VHF Radio and Automatic Identification Systems (AIS at Sea), which was published in 2006.

SECTION 2 – ANALYSIS

2.1 AIM

The purpose of the analysis is to determine the contributory causes and circumstances of the accident as a basis for making recommendations to prevent similar accidents occurring in the future.

2.2 THE COLLISION

The collision between *Ever Smart* and *Alexandra 1* stemmed from the vessels' masters having differing perceptions of how the vessels would pass each other. The tanker's master assumed that the container ship would alter to port on leaving the buoyed channel and pass astern. Consequently, as *Ever Smart* neared the No1 buoys, he thought it safe to manoeuvre *Alexandra 1* slowly ahead across the channel entrance. The container ship's master, however, assumed that the tanker was stationary and would pass down his port side. Therefore, after the pilot had disembarked, he remained on the same heading and increased speed. The actions of both masters put their vessels on a steady bearing (**Figure 8**).

Within 1 minute of *Ever Smart* leaving the channel, *Alexandra 1*'s master noticed that the container ship had not altered course as he had expected. He then acted immediately to try and prevent a collision. However, *Ever Smart*'s master was not monitoring *Alexandra 1*. Therefore, he was not aware of its proximity until alerted by VHF calls seconds before the collision. By then, it was too late for him to take any effective avoiding action. Unfortunately, the setting of 'full astern' on board *Alexandra 1* was also too late to prevent the collision (**Figures 10 and 11**).

The reliance of *Alexandra 1*'s master on scanty VHF information and the failure of *Ever Smart*'s master to keep a proper lookout and monitor *Alexandra 1*'s movement were pivotal to this accident. However, it is also evident that a lack of an agreed plan and effective communication, co-ordination and monitoring were significant factors, which contributed to the flaws in *Ever Smart*'s and *Alexandra 1*'s masters' situational awareness. On this occasion, the precautions of pilotage and VTS in Jebel Ali, which should have been able to manage and de-conflict the vessels' movements, were ineffective.

2.3 VESSEL PERSPECTIVES

2.3.1 *Alexandra 1*

The VTSO instructed *Alexandra 1* to be at No1 buoys at 2315 (**Table 1**). The VTSO also authorised the master to enter the channel as soon as the container ship was clear (**Table 3**). It is evident from *Alexandra 1*'s master's handover notes (paragraph 1.5.2) that he was used to having to remain at least 1.5nm from the No1 buoys when waiting for a pilot in Jebel Ali. Consequently, although on this occasion the VTSO did not instruct *Alexandra 1* to wait a specific distance from the buoys, the master's decision to stop the vessel's engine 1.3nm from the channel at 2314 indicates that he was following usual practice.

However, at this point, *Ever Smart* had only just passed No8 buoys and, over the following 25 minutes, while *Alexandra 1* waited for the container ship to clear the channel, the tanker closed the channel entrance due to a combination of 'dead slow ahead' engine movements and drift. No attempt was made to check *Alexandra 1*'s movement to the east.

By itself, *Alexandra 1*'s proximity to the channel entrance was not hazardous providing the masters of both vessels were aware of the other's intentions. As soon as the tanker's master mistook the VTSO's instruction to '*pass astern of the tanker*' as an instruction to *Ever Smart* rather than *Zakheer Bravo*, his perception of how the vessels would pass differed dramatically from that of *Ever Smart*'s master.

Until 2329, *Alexandra 1* had remained to the north-west of the channel and had not crossed the line of the channel's entrance (**Figures 3, 4 and 6**). However, after *Alexandra 1*'s master assessed that *Ever Smart* would need to alter course to port on leaving the channel in order to pass astern, he manoeuvred the tanker slowly ahead. This was probably done to give *Ever Smart* more sea room and to enable the approaching pilot boat to come alongside. The tanker's master was not aware that the container ship's master did not intend to alter course. He also did not appreciate the danger of straddling the slow moving and relatively un-maneuvrable *Alexandra 1* across the line of the buoyed channel just 4 cables from its entrance.

2.3.2 *Ever Smart*

When the pilot informed *Ever Smart*'s master that he would be disembarking, the master's question '*do you think I can go myself?*' (**Table 5**) indicated that he was a little surprised that the pilot was not staying on board until the containership had left the channel. Nonetheless, he did not challenge the pilot's departure.

At the time, *Ever Smart* was following the channel as intended, and it was only 1.3nm from the channel end. *Alexandra 1* was 2.9nm off the container ship's port bow and the tanker was not impeding *Ever Smart*'s passage to seaward. Moreover, the pilot had informed the master that the tanker would wait for *Ever Smart* to clear the channel. Therefore, it was reasonable for the master to assume that the tanker would pass down the container ship's port side. At no time did the master or pilot discuss the option of altering course to leave *Alexandra 1* on the container vessel's starboard side.

2.4 VHF COMMUNICATIONS WATCH

In busy port areas, the clarity and accuracy of VHF traffic is essential. However, the lack of discipline on VHF radio is a common problem in some regions. Constant 'chatter' resulting in exchanges being over-spoken is a regular occurrence. Consequently, the possibility of missing transmissions or parts of transmissions is increased.

In this case, the port's working channel, VHF channel 69, was very busy and *Alexandra 1*'s master did not hear the full exchange between port control and the tug *Zakheer Bravo*. As a result, although the master correctly assumed that the 'tanker' referred to in the exchange was *Alexandra 1*, he clearly did not know the name of the vessel the VTSO was talking to. In such circumstances, it would have been appropriate for the master to clarify the situation with either port control or *Ever Smart* rather than taking action on the basis of incomplete information.

Jebel Ali port control had sufficient personnel to provide a 24 hour service and it was equipped to monitor and communicate with vessels in the port area. The port's operational procedures were also comprehensive and had been revised to reflect the lessons learned from the collision between *Kashmir* and *Sima Saman* in 2009 (paragraph 1.6.7). Nonetheless, although the VTSOs completed on the job training and were to be closely supervised, very few held a V103 qualification. In addition, the VTSOs did not participate in emergency drills. As a result, the VTSOs might not have been adequately equipped to recognise when potentially hazardous situations were developing and how to respond accordingly.

2.10 THE USE OF AIS

Alexandra 1's AIS was not operating. Therefore, information such as the vessel's name was not readily available to *Ever Smart*'s bridge team or the VTSO. Similarly, AIS information transmitted from *Ever Smart* and other vessels such as *Zakheer Bravo* was not available on board *Alexandra 1*. The lack of AIS information possibly contributed to the tanker master's ignorance of the container ship's identity immediately before the collision. It also possibly contributed to the container ship master's ignorance of the tanker's identity immediately after. However, the names of the vessels had been mentioned on the VHF radio on numerous occasions before the collision. Therefore, AIS was not the only source of such information.

AIS must be used with caution and the reliance placed on the system for collision avoidance by *Ever Smart*'s bridge watchkeepers, which is becoming increasingly widespread, is of concern. AIS has certain advantages over ARPA (paragraph 1.8.2) and, except for security reasons or specific exemptions, the system should always be operated on board ships on which it is required to be carried.

In this case, there appears to be no reason why *Alexandra 1*, which was entering a large commercial port, had its AIS turned off. Therefore, it is surprising that Jebel Ali port control had not challenged *Alexandra 1* over its lack of AIS transmissions during this and the vessel's previous visits, particularly as AIS was its preferred and most reliable method of tracking vessels using the port.

SECTION 3 – CONCLUSIONS

3.1 SAFETY ISSUES DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. The reliance of *Alexandra 1*'s master on scanty VHF information and the failure of *Ever Smart*'s master to keep a proper lookout and monitor *Alexandra 1*'s movement were pivotal to this accident. [2.2, 2.4, 2.5]
2. The precautions of pilotage and VTS in Jebel Ali, which should have been able to manage and de-conflict the vessels' movements, were ineffective. [2.2]
3. The VTSO's instructions to *Alexandra 1* resulted in the tanker arriving off the buoyed channel 25 minutes earlier than was necessary. While the tanker waited for *Ever Smart* to clear the channel, no action was taken to stop the tanker from slowly moving towards the channel's entrance. [2.3.1, 2.7.1]
4. The movement of *Alexandra 1* into *Ever Smart*'s path resulted from its master's assumption that *Ever Smart* would turn to port on leaving the channel. [2.3.1]
5. *Alexandra 1*'s master did not appreciate the danger of straddling the slow-moving, relatively un-maneuvrable tanker across the line of the buoyed channel just 4 cables from its entrance. [2.3.1]
6. During the 8 minutes from the pilot's disembarkation until the collision, *Ever Smart*'s bridge team did not monitor *Alexandra 1*'s position and movement. Both the master and the third officer lost situational awareness. [2.5]
7. The decision of *Alexandra 1*'s master to call Jebel Ali port control, rather than calling *Ever Smart* when he realised the container ship had not altered course, cost valuable seconds. [2.6]
8. The VTSO did not challenge *Alexandra 1*'s proximity to the channel entrance or appreciate the seriousness of the situation when he was alerted by *Alexandra 1*'s master. [2.7.2]
9. The VTSO's VHF exchanges with *Alexandra 1* and *Ever Smart* immediately before the collision indicate that he did not know how to communicate effectively in an emergency. [2.7.3]
10. The pilot monitored *Alexandra 1* but it is apparent that he had not appreciated how close the tanker was to the channel entrance or its potential closest point of approach with *Ever Smart*. [2.8.1]
11. The pilot's failure to co-ordinate and communicate the passing arrangements for *Ever Smart* and *Alexandra 1* was significant. [2.8.2]
12. The pilot's disembarkation from *Ever Smart* was premature. [2.8.3]

3.2 SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT THAT HAVE BEEN ADDRESSED OR RESULTED IN RECOMMENDATIONS

1. Most of the port's VTSOs had not completed the IALA V103 qualification or participated in emergency drills. Therefore they might not have been adequately equipped to recognise when potentially hazardous situations were developing and how to respond accordingly. [2.9]
2. The AIS on board *Alexandra 1* was not operating. This was contrary to the requirements of SOLAS and meant that information such as vessel names was not readily available on board *Alexandra 1* and *Ever Smart*, or to the VTSO. [2.10]

3.3 OTHER SAFETY ISSUES NOT DIRECTLY CONTRIBUTING TO THE ACCIDENT¹⁰

1. *Alexandra 1*'s master had doubts regarding the actions of *Ever Smart*, but no sound signals were made. [2.6]

¹⁰ These safety issues identify lessons to be learned. They do not merit a safety recommendation based on this investigation alone. However, they may be used for analysing trends in marine accidents or in support of a future safety recommendation